

# GALIB RAHAMAN MALLIK

✉ [galibmallik3@gmail.com](mailto:galibmallik3@gmail.com) | ☎ +91-9769586285 | 📍 Bengaluru, India

## PROFILE

Generative AI Engineer & Senior Data Scientist with 7+ years of academic and industry experience in AI/ML, specializing in **Large Reasoning Models (LRMs)**, and **Multi-Agent AI (LangGraph, CrewAI, Autogen)**, **Generative AI (GANs, VAEs, Diffusion, Flow Matching)**, **Retrieval-Augmented Generation (RAG)**. Strong expertise in **fine-tuning (LoRA, PEFT)**, **reinforcement learning (PPO, GRPO)**, and applying AI in domains like **scalable enterprise AI solutions across healthcare, bioinformatics, ERP automation, and robotics**. Proven track record in delivering AI-driven business solutions and leading innovation projects.

## EXPERIENCE

### Senior Data Scientist – Fujitsu Consulting (2021–Present)

Bengaluru, India

- Designed and deployed Agentic AI frameworks with LangGraph, CrewAI, Autogen.
- Built MCP servers & clients enabling AI agents for reasoning and automation
- Led Generative AI projects: protein backbone generation, medical RAG system, SQL generation from natural language (Oracle cloud).
- Developed Fine-Tuned LLMs (GPT, Qwen, Llama, Gemini) using DPO, LoRA, GRPO.
- Authored whitepapers and presented AI use cases internally.

### Consultant – Tata Advanced Systems (2019–2021)

Bengaluru, India

- Applied AI in aerospace/robotics: object detection, tracking, 3D reconstruction.
- Delivered face recognition & navigation systems for security/defense applications

### Research Fellow – IIT Bombay & Kansas State Univ. (2017–2019)

India & USA

- Developed multi-agent algorithms for autonomous systems.
- Published at top conferences (Indian Control Conf., European Control Conf., IFAC).
- Mentored students, contributed to robotics navigation & AI-based stability analysis.

## KEY PROJECTS

### Agent AI chatbot using MCP server & client (Agentic AI) - 2025

- MCP server/client with LangGraph & Claude Desktop
- Integration of third party MCP servers
- Development of custom MCP server

### Development of Reasoning Language model (Fine Tuned) - 2025

- Fine-tuning GPT/Qwen with DPO & GRPO.
- Development of preferential data set
- PEFT with LORA

### U-Oracle GenAI (Oracle Cloud interface) - 2025

- Generation of SQL query from natural language & Fit gap analysis of ERP documents
- LLM: Cohere command R+
- Oracle financial fusion module (Fusion database)

### Common Dictionary Standardization (Agentic AI) - 2025

- Agentic RAG with langgraph

- Implemented Self-RAG: Langgraph
- LLM & Embedding: GPT-4o & Hugging Face OpenAI (Chat API)

### **Protein backbone generation (Generative AI) - 2024**

- Generative AI with SE(3) Lie groups, diffusion & flow matching.

### **Nurbugring (Computer vision) - 2024**

- YOLOv9 with advanced object tracking
- Implemented tracking algorithms on the detected objects (bounding boxes): ByteTrack, Deep Sort algorithms

### **Medical RAG System (Generative AI) - 2024**

- GPT-based RAG integrating clinical data
- Prompt Engineering
- Evaluation of RAG system

### **Employee Experience (AutoML) -2023**

- AutoM: automating implementation of data analysis tasks
- Using SapientML, developed the prototype with sample data
- Built web interface using Streamlit

## **SKILLS**

- **Programing Language:** Python, C++, Matlab
- **LLMs & Agents:** GPT, Qwen, Llama, Gemini, LangChain, LangGraph, CrewAI, AutoGen, MCP
- **Generative AI:** GANs, VAEs, Diffusion, Flow Matching, Protein AI
- **Frameworks:** Huggingface, Pytorch, Pytorch Geometric, TensorFlow, LangSmith, Unsloth
- **Tools:** FastAPI, Flask, MySQL, Docker, Streamlit, Gradio,

## **EDUCATION**

- **Ph.D., Systems & Control Engineering, IIT Bombay (2017)** – Multi-agent formation control, data-driven algorithms, vision based navigation
- **M.E., Control Systems Engineering, Jadavpur Univ. (2008)** – Laser Beam Stabilization.
- **AMIEETE (Equivalent B.Tech), IETE (2003)** – Electronics & Telecom Engineering

## **PUBLICATIONS & AWARDS**

- Geometric Deep Learning: Framework for Complex Data – Fujitsu Convention (2024).
- Generative AI: A Comprehensive Study – under preparation
- Multiple research papers in European Journal of Control, ECC, ICC, IFAC.
- Topper in M.E. (Jadavpur University).
- Qualified GATE (2006).

## **SUMMARY**

A results-driven AI researcher and practitioner with strong expertise in **Generative AI, LLMs, and Multi-Agent Systems**. Blends deep research with enterprise-scale AI deployment across industries: enterprise AI, robotics, aerospace, healthcare and bioinformatics.